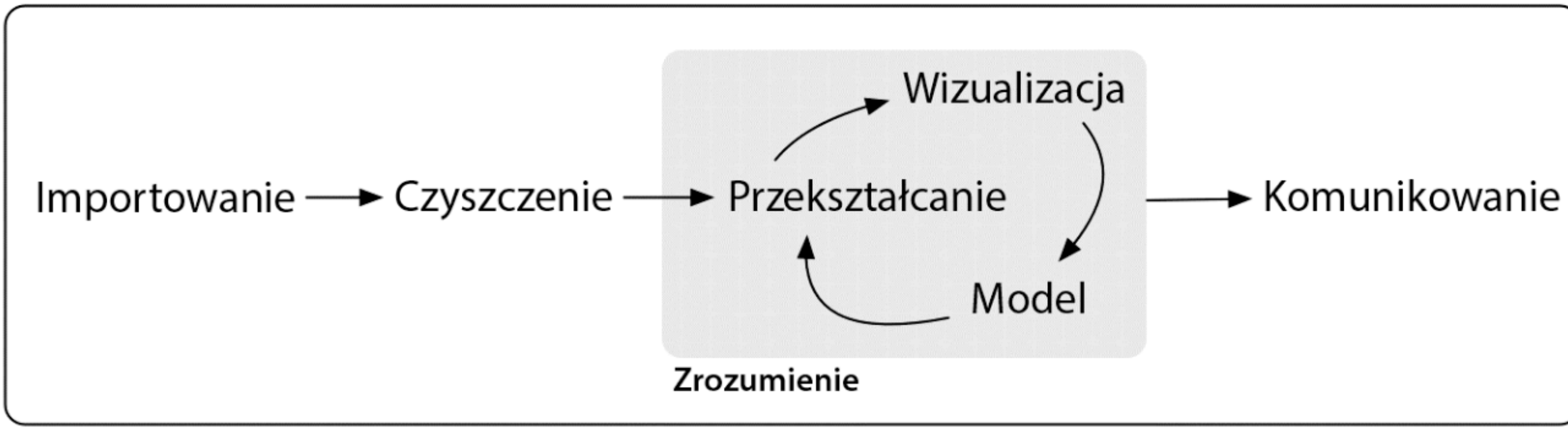
Podstawy gramatyki graficznej z ggplot

slajdy pomocnicze kurs IDUB

Tomasz Olczyk 12.06.2025

Czego się nauczymy?

- podstaw teorii gramatyki graficznej
- implementacji teorii gramatyki graficznej w ggplot
- struktury i procedury tworzenia wykresów w ggplot
- działania funkcji ggplot()
- rzutowania (mapowania) danych na obiekty graficzne funkcją aes( )
- działania funkcji obiektów graficznych geom_*( )
- działania funkcji generujących współrzędne coord_*( )
- rysowania map z wykorzystaniem geom_polygon( ) i funkcji
- tworzenia wykresów małych wielokrotności funkcjami facet_grid i facet_wrap
- dodawania etykiet funkcją labs( )
- zmienienia motywu wykresu funkcjami theme_*


Importowanie
Czyszczenie
Przekształcanie
Wizualizacja
Model
Komunikowanie
Zrozumienie

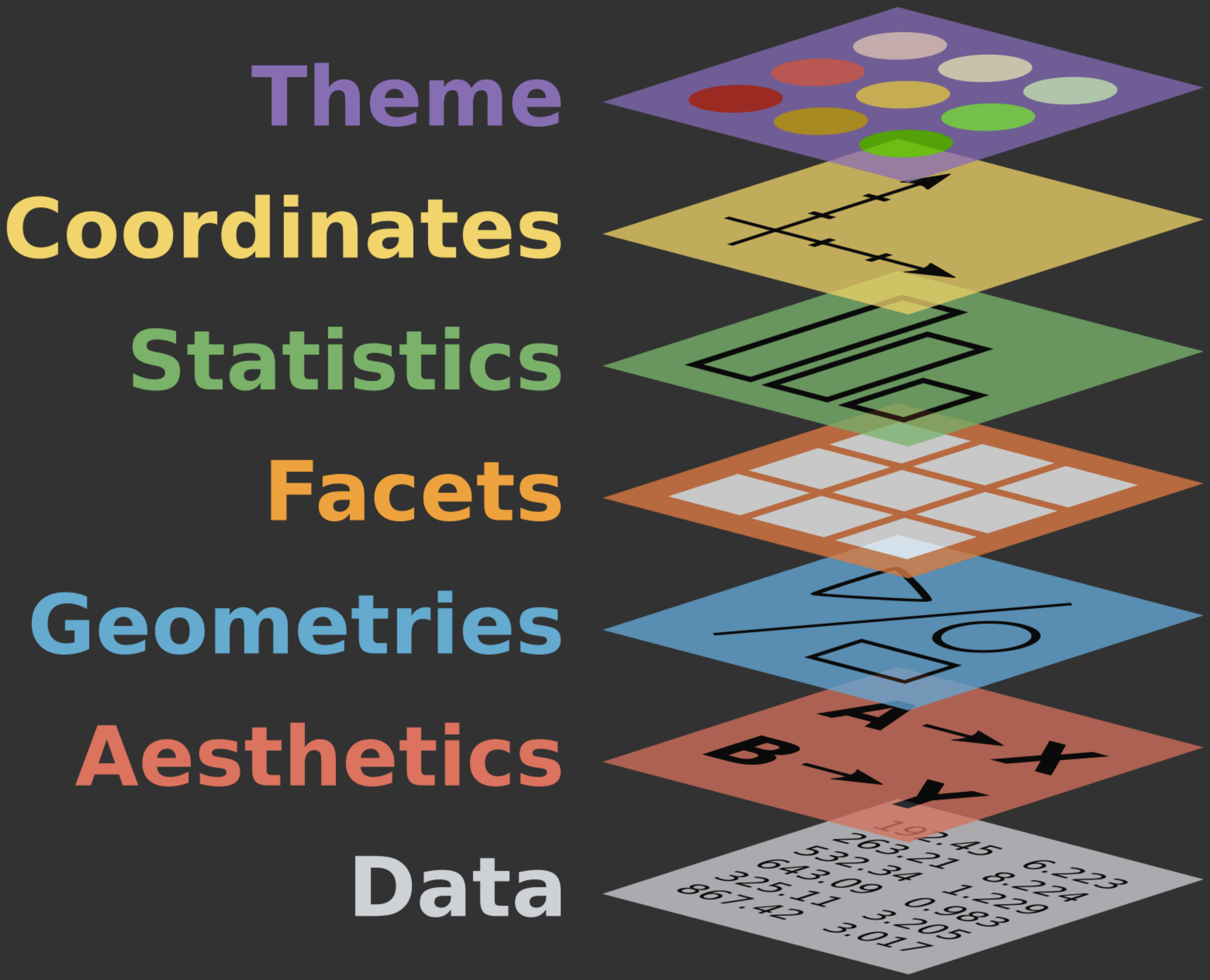
Statistics and Computing

Leland Wilkinson

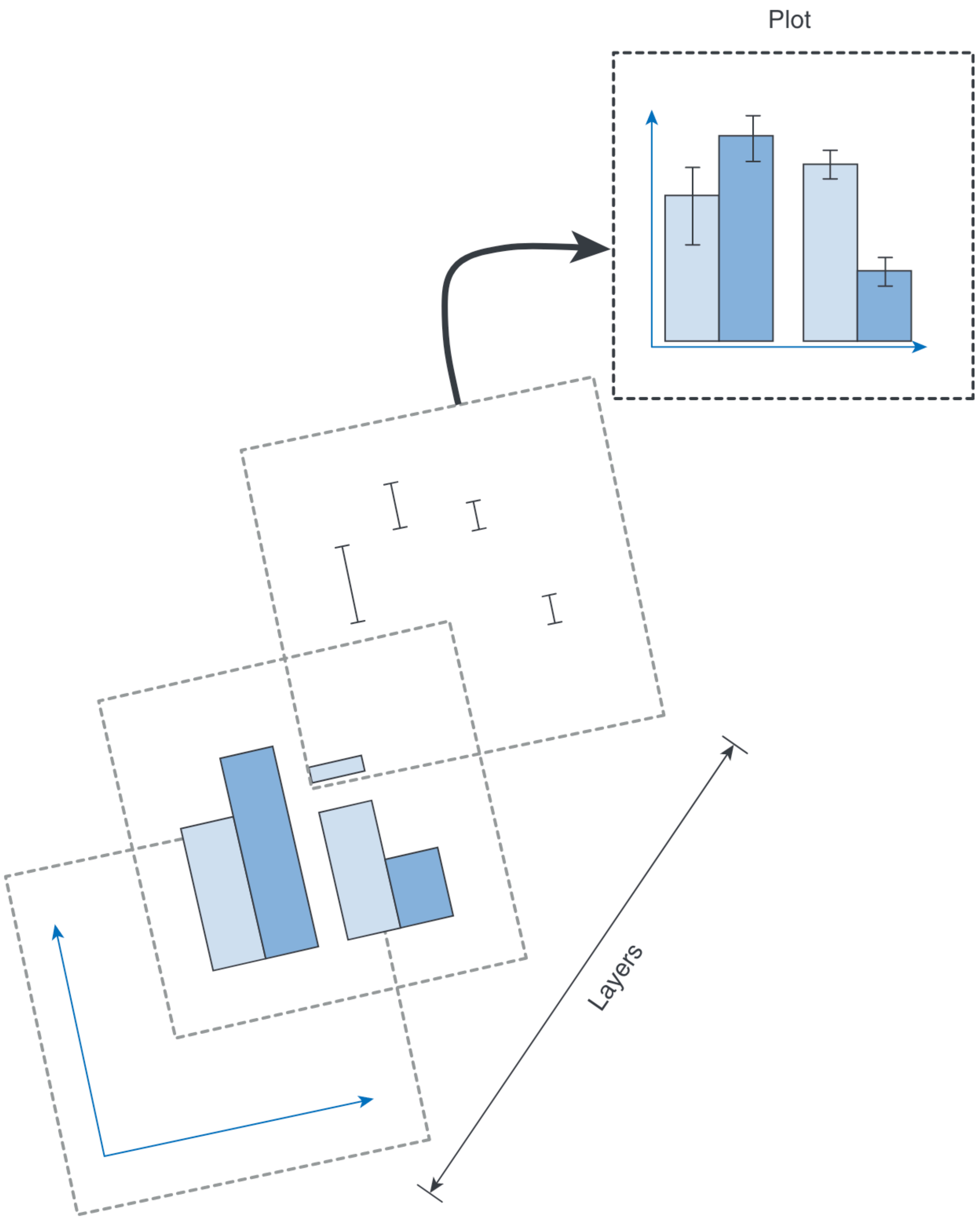
The Grammar of Graphics

Second Edition

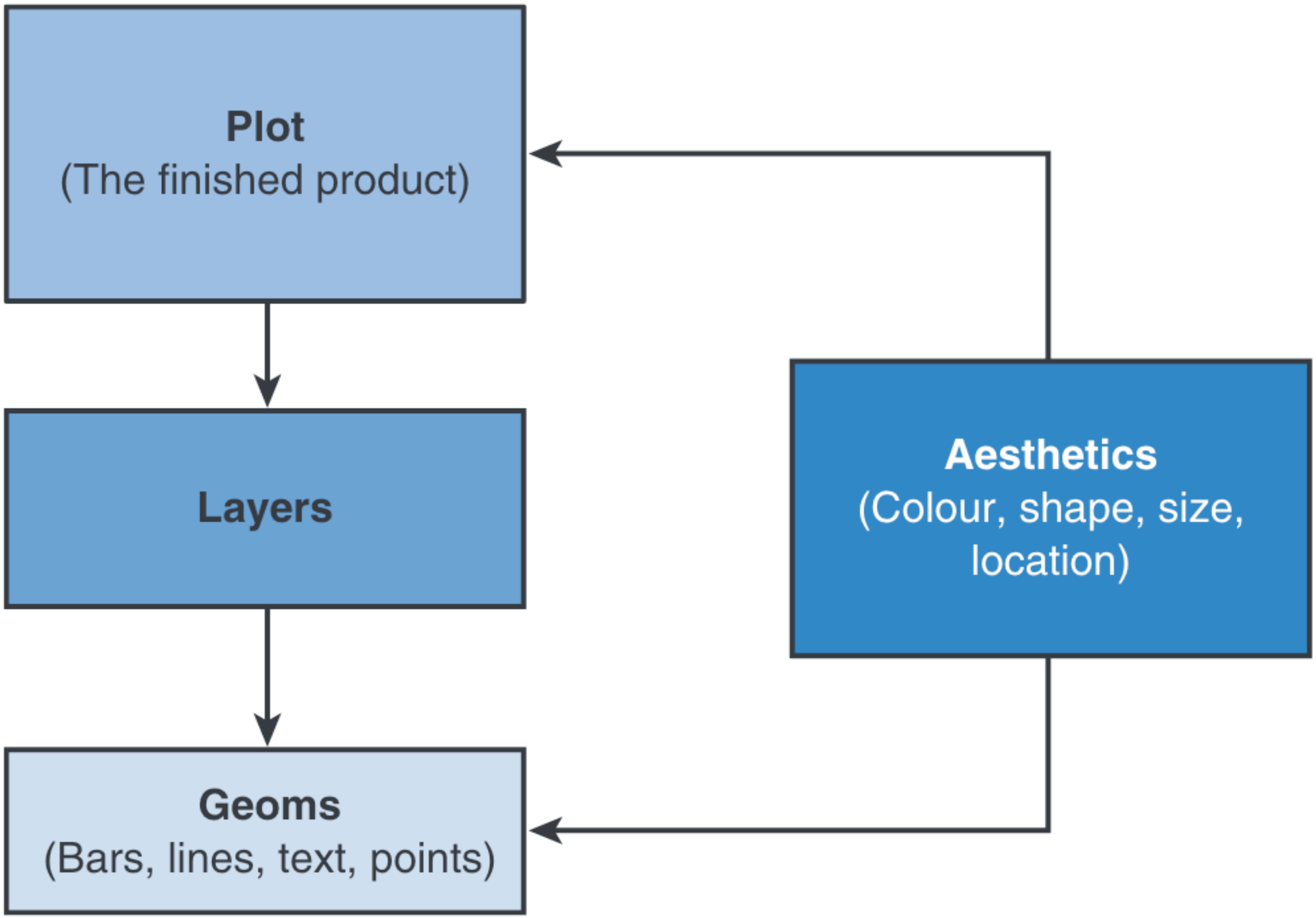
Springer

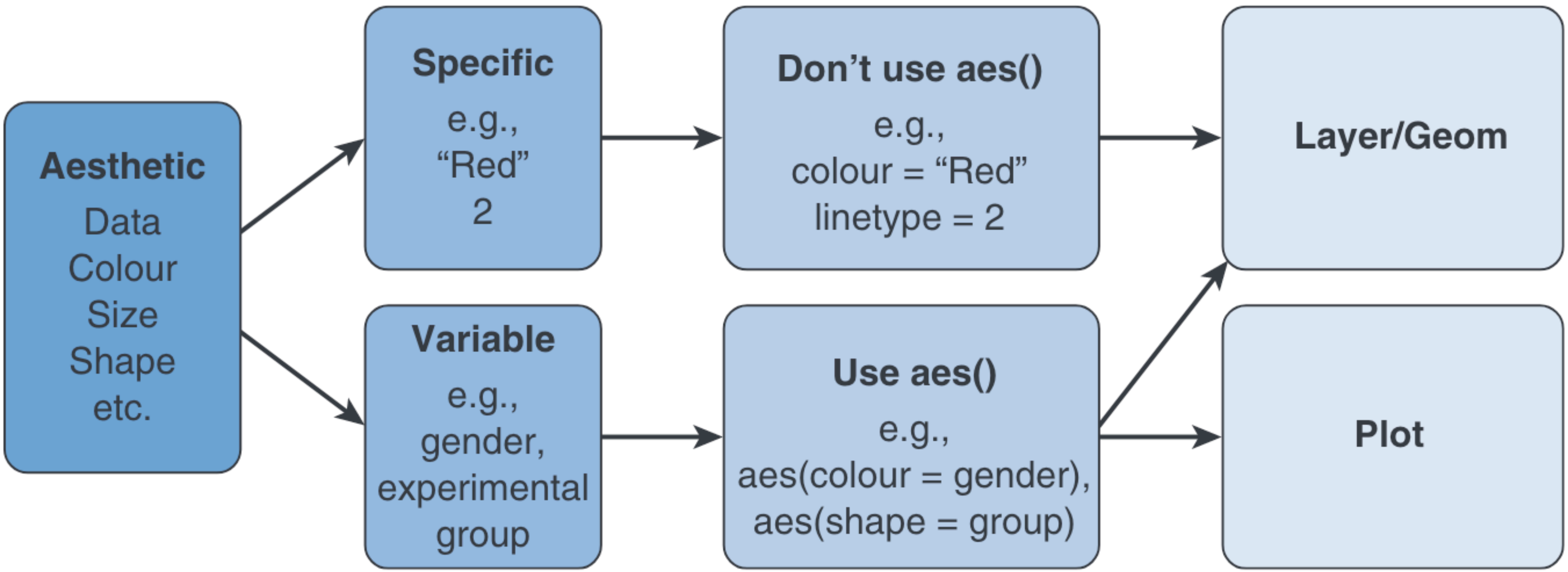

Theme
Coordinates
Statistics
Facets
Geometries
Aesthetics
Data
A
B
X
Y
192.45
263.21
532.34
643.09
325.11
867.42
6.223
8.224
1.229
0.983
3.205
3.017

```
ggplot(data = <DANE>) +
   <FUNKCJA_GEOMETRYCZNA>(
     mapping = aes(<MAPOWANIA>),
      stat = <PRZEKSZTAŁCENIE_STATYSTYCZNE>,
      position = <POZYCJA>
  )+
  <FUNKCJA_WSPÓŁRZĘDNYCH> +
  <FUNKCJA_PANELI>
```



Plot
Layers


Plot
(The finished product)
Layers
Geoms
(Bars, lines, text, points)
Aesthetics
(Colour, shape, size,
location)


Aesthetic
Data
Colour
Size
Shape
etc.
Specific
e.g.,
“Red”
2
Variable
e.g.,
gender,
experimental
group
Don’t use aes()
e.g.,
colour = “Red”
linetype = 2
Use aes()
e.g.,
aes(colour = gender),
aes(shape = group)
Layer/Geom
Plot

	Required	Optional
geom_bar()	x: the variable to plot on the *x*-axis	colour size fill linetype weight alpha
geom_point()	x: the variable to plot on the *x*-axis y: the variable to plot on the *y*-axis	shape colour size fill alpha
geom_line()	x: the variable to plot on the *x*-axis y: the variable to plot on the *y*-axis	colour size linetype alpha
geom_smooth()	x: the variable to plot on the *x*-axis y: the variable to plot on the *y*-axis	colour size fill linetype weight alpha
geom_histogram()	x: the variable to plot on the *x*-axis	colour size fill linetype weight alpha
geom_boxplot()	x: the variable to plot ymin: lower limit of 'whisker' ymax: upper limit of 'whisker' lower: lower limit of the 'box' upper: upper limit of the 'box' middle: the median	colour size fill weight alpha
geom_text()	x: the horizontal coordinate of where the text should be placed y: the vertical coordinate of where the text should be placed label: the text to be printed all of these can be single values or variables containing coordinates and labels for multiple items	colour size angle hjust (horizontal adjustment) vjust (vertical adjustment) alpha
geom_density()	x: the variable to plot on the *x*-axis y: the variable to plot on the *y*-axis	colour size fill linetype weight alpha

Aesthetic	Option	Outcome
Linetype	linetype = 1	Solid line (default)
	linetype = 2	Hashed
	linetype = 3	Dotted
	linetype = 4	Dot and hash
	linetype = 5	Long hash
	linetype = 6	Dot and long hash
Size	size = value	Replace 'value' with a value in mm (default size = 0.5). Larger values than 0.5 give you fatter lines/larger text/bigger points than the default whereas smaller values will produce thinner lines/smaller text and points than the default.
	e.g., size = 0.25	Produces lines/points/text of 0.25mm
Shape	shape = integer, shape = "x"	The integer is a value between 0 and 25, each of which specifies a particular shape. Some common examples are below. Alternatively, specify a single character in quotes to use that character (shape = "A" will plot each point as the letter A).
	shape = 0	Hollow square (15 is a filled square)
	shape = 1	Hollow circle (16 is a filled circle)
	shape = 2	Hollow triangle (17 is a filled triangle)
	shape = 3	'+'
	shape = 5	Hollow rhombus (18 for filled)
	shape = 6	Hollow inverted triangle
Colour	colour = "Name"	Simply type the name of a standard colour. For example, colour = "Red" will make the geom red.
	colour = "#RRGGBB"	Specify exact colours using the RRGGBB system. For example, colour = "#3366FF" produces a shade of blue, whereas colour = "#336633" produces a dark green.
Alpha	alpha(colour, value)	Colours can be made transparent by specifying alpha, which can range from 0 (fully transparent) to 1 (fully opaque). For example, alpha("Red", 0.5) will produce a half transparent red.

#1

white
aliceblue
antiquewhite
antiquewhite1
antiquewhite2
antiquewhite3
antiquewhite4
aquamarine
aquamarine1
aquamarine2
aquamarine3
aquamarine4
azure
azure1
azure2
azure3
azure4
beige
bisque
bisque1

bisque2
bisque3
bisque4
black
blanchedalmond
blue
blue1
blue2
blue3
blue4
blueviolet
brown
brown1
brown2
brown3
brown4
burlywood
burlywood1
burlywood2
burlywood3

burlywood4
cadetblue
cadetblue1
cadetblue2
cadetblue3
cadetblue4
chartreuse
chartreuse1
chartreuse2
chartreuse3
chartreuse4
chocolate
chocolate1
chocolate2
chocolate3
chocolate4
coral
coral1
coral2
coral3

coral4
cornflowerblue
cornsilk
cornsilk1
cornsilk2
cornsilk3
cornsilk4
cyan
cyan1
cyan2
cyan3
cyan4
darkblue
darkcyan
darkgoldenrod
darkgoldenrod1
darkgoldenrod2
darkgoldenrod3
darkgoldenrod4
darkgray

darkgreen
darkgrey
darkkhaki
darkmagenta
darkolivegreen
darkolivegreen1
darkolivegreen2
darkolivegreen3
darkolivegreen4
darkorange
darkorange1
darkorange2
darkorange3
darkorange4
darkorchid
darkorchid1
darkorchid2
darkorchid3
darkorchid4
darkred

#2

darksalmon	deepskyblue	ghostwhite	gray8	gray28
darkseagreen	deepskyblue1	gold	gray9	gray29
darkseagreen1	deepskyblue2	gold1	gray10	gray30
darkseagreen2	deepskyblue3	gold2	gray11	gray31
darkseagreen3	deepskyblue4	gold3	gray12	gray32
darkseagreen4	dimgray	gold4	gray13	gray33
darkslateblue	dimgrey	goldenrod	gray14	gray34
darkslategray	dodgerblue	goldenrod1	gray15	gray35
darkslategray1	dodgerblue1	goldenrod2	gray16	gray36
darkslategray2	dodgerblue2	goldenrod3	gray17	gray37
darkslategray3	dodgerblue3	goldenrod4	gray18	gray38
darkslategray4	dodgerblue4	gray	gray19	gray39
darkslategrey	firebrick	gray0	gray20	gray40
darkturquoise	firebrick1	gray1	gray21	gray41
darkviolet	firebrick2	gray2	gray22	gray42
deeppink	firebrick3	gray3	gray23	gray43
deeppink1	firebrick4	gray4	gray24	gray44
deeppink2	floralwhite	gray5	gray25	gray45
deeppink3	forestgreen	gray6	gray26	gray46
deeppink4	gainsboro	gray7	gray27	gray47

#3

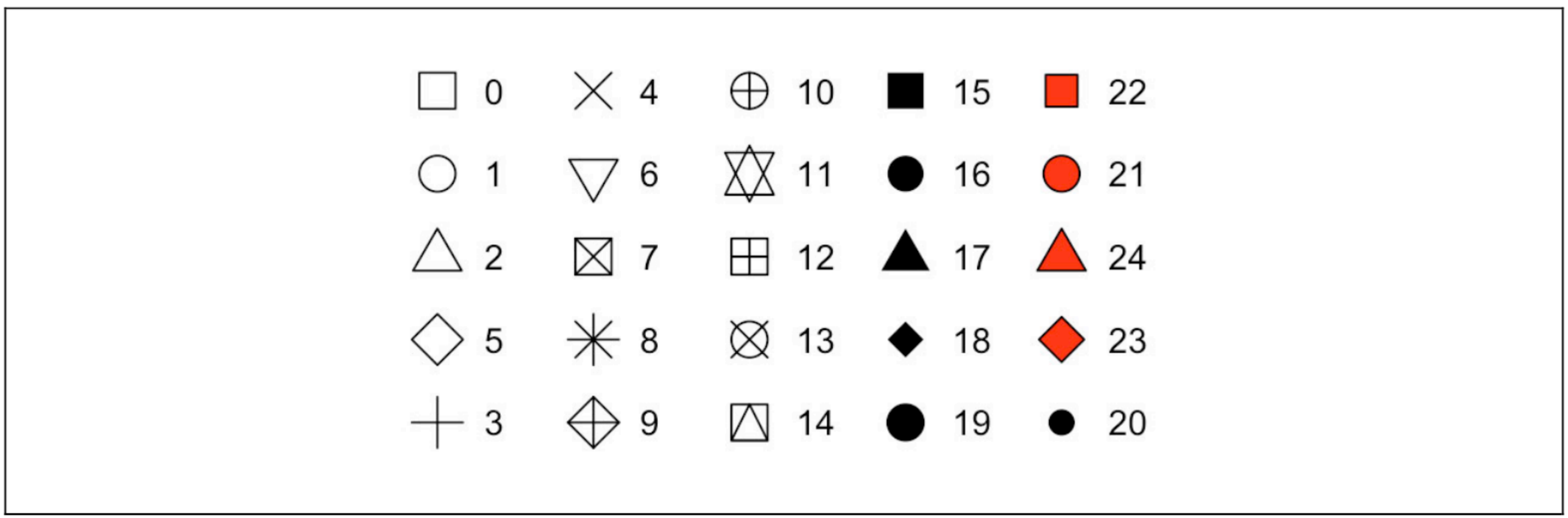
gray48	gray68	gray88	grey0	grey20
gray49	gray69	gray89	grey1	grey21
gray50	gray70	gray90	grey2	grey22
gray51	gray71	gray91	grey3	grey23
gray52	gray72	gray92	grey4	grey24
gray53	gray73	gray93	grey5	grey25
gray54	gray74	gray94	grey6	grey26
gray55	gray75	gray95	grey7	grey27
gray56	gray76	gray96	grey8	grey28
gray57	gray77	gray97	grey9	grey29
gray58	gray78	gray98	grey10	grey30
gray59	gray79	gray99	grey11	grey31
gray60	gray80	gray100	grey12	grey32
gray61	gray81	green	grey13	grey33
gray62	gray82	green1	grey14	grey34
gray63	gray83	green2	grey15	grey35
gray64	gray84	green3	grey16	grey36
gray65	gray85	green4	grey17	grey37
gray66	gray86	greenyellow	grey18	grey38
gray67	gray87	grey	grey19	grey39

#4

grey40	grey60	grey80	grey100	ivory4
grey41	grey61	grey81	honeydew	khaki
grey42	grey62	grey82	honeydew1	khaki1
grey43	grey63	grey83	honeydew2	khaki2
grey44	grey64	grey84	honeydew3	khaki3
grey45	grey65	grey85	honeydew4	khaki4
grey46	grey66	grey86	hotpink	lavender
grey47	grey67	grey87	hotpink1	lavenderblush
grey48	grey68	grey88	hotpink2	lavenderblush1
grey49	grey69	grey89	hotpink3	lavenderblush2
grey50	grey70	grey90	hotpink4	lavenderblush3
grey51	grey71	grey91	indianred	lavenderblush4
grey52	grey72	grey92	indianred1	lawngreen
grey53	grey73	grey93	indianred2	lemonchiffon
grey54	grey74	grey94	indianred3	lemonchiffon1
grey55	grey75	grey95	indianred4	lemonchiffon2
grey56	grey76	grey96	ivory	lemonchiffon3
grey57	grey77	grey97	ivory1	lemonchiffon4
grey58	grey78	grey98	ivory2	lightblue
grey59	grey79	grey99	ivory3	lightblue1

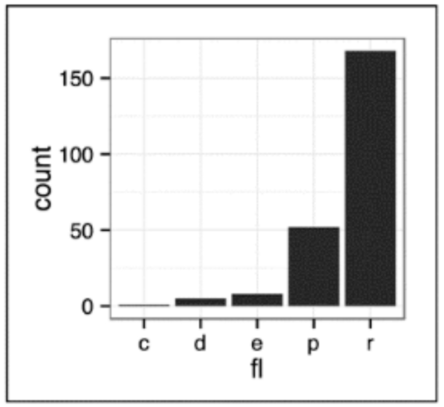
#5

lightblue2
lightblue3
lightblue4
lightcoral
lightcyan
lightcyan1
lightcyan2
lightcyan3
lightcyan4
lightgoldenrod
lightgoldenrod1
lightgoldenrod2
lightgoldenrod3
lightgoldenrod4
lightgoldenrodyellow
lightgray
lightgreen
lightgrey
lightpink
lightpink1
lightpink2
lightpink3
lightpink4
lightsalmon
lightsalmon1
lightsalmon2
lightsalmon3
lightsalmon4
lightseagreen
lightskyblue
lightskyblue1
lightskyblue2
lightskyblue3
lightskyblue4
lightslateblue
lightslategray
lightslategrey
lightsteelblue
lightsteelblue1
lightsteelblue2
lightsteelblue3
lightsteelblue4
lightyellow
lightyellow1
lightyellow2
lightyellow3
lightyellow4
limegreen
linen
magenta
magenta1
magenta2
magenta3
magenta4
maroon
maroon1
maroon2
maroon3
maroon4
mediumaquamarine
mediumblue
mediumorchid
mediumorchid1
mediumorchid2
mediumorchid3
mediumorchid4
mediumpurple
mediumpurple1
mediumpurple2
mediumpurple3
mediumpurple4
mediumseagreen
mediumslateblue
mediumspringgreen
mediumturquoise
mediumvioletred
midnightblue
mintcream
mistyrose
mistyrose1
mistyrose2
mistyrose3
mistyrose4
moccasin
navajowhite
navajowhite1
navajowhite2
navajowhite3
navajowhite4
navy
navyblue
oldlace
olivedrab
olivedrab1
olivedrab2
olivedrab3
olivedrab4
orange
orange1
orange2

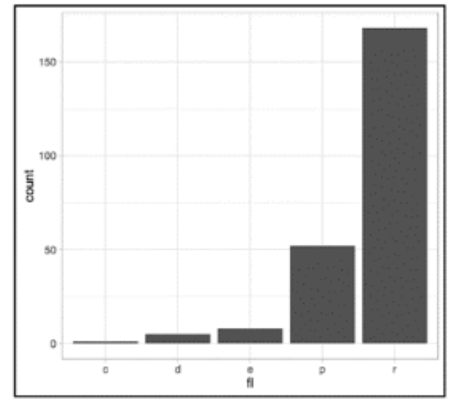

0
4
10
15
22
1
6
11
16
21
2
7
12
17
24
5
8
13
18
23
3
9
14
19
20

Szablony

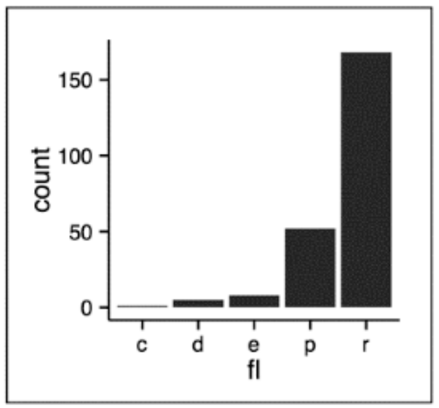
Funkcje szablonów zmieniają wygląd wykresu



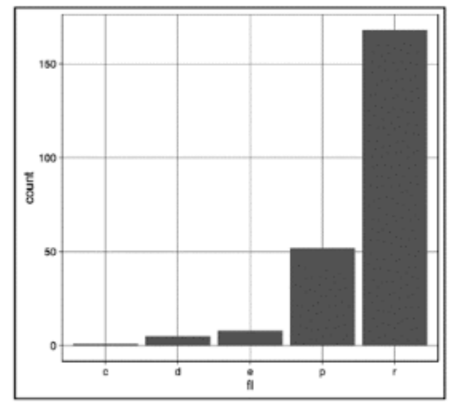
theme_bw()
Białe tło
z liniami siatki



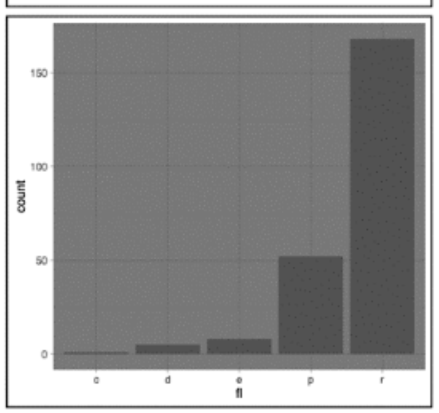
theme_light()
Jasne osie
i linie siatki



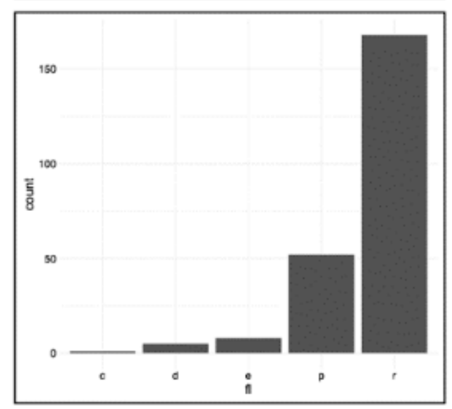
theme_classic()
Klasyczny szablon,
osie, bez linii siatki



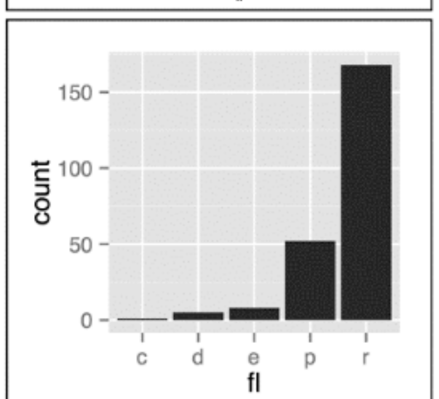
theme_linedraw()
Tylko czarne linie



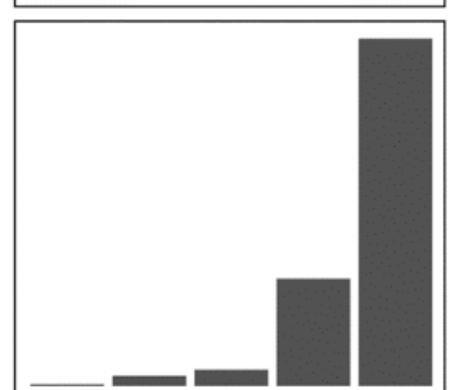
theme_dark()
Ciemne tło
tworzące kontrast



theme_minimal()
Minimalny szablon,
widoczne są
tylko figury
geometryczne



theme_gray()
Szare tło
(szablon domyślny)

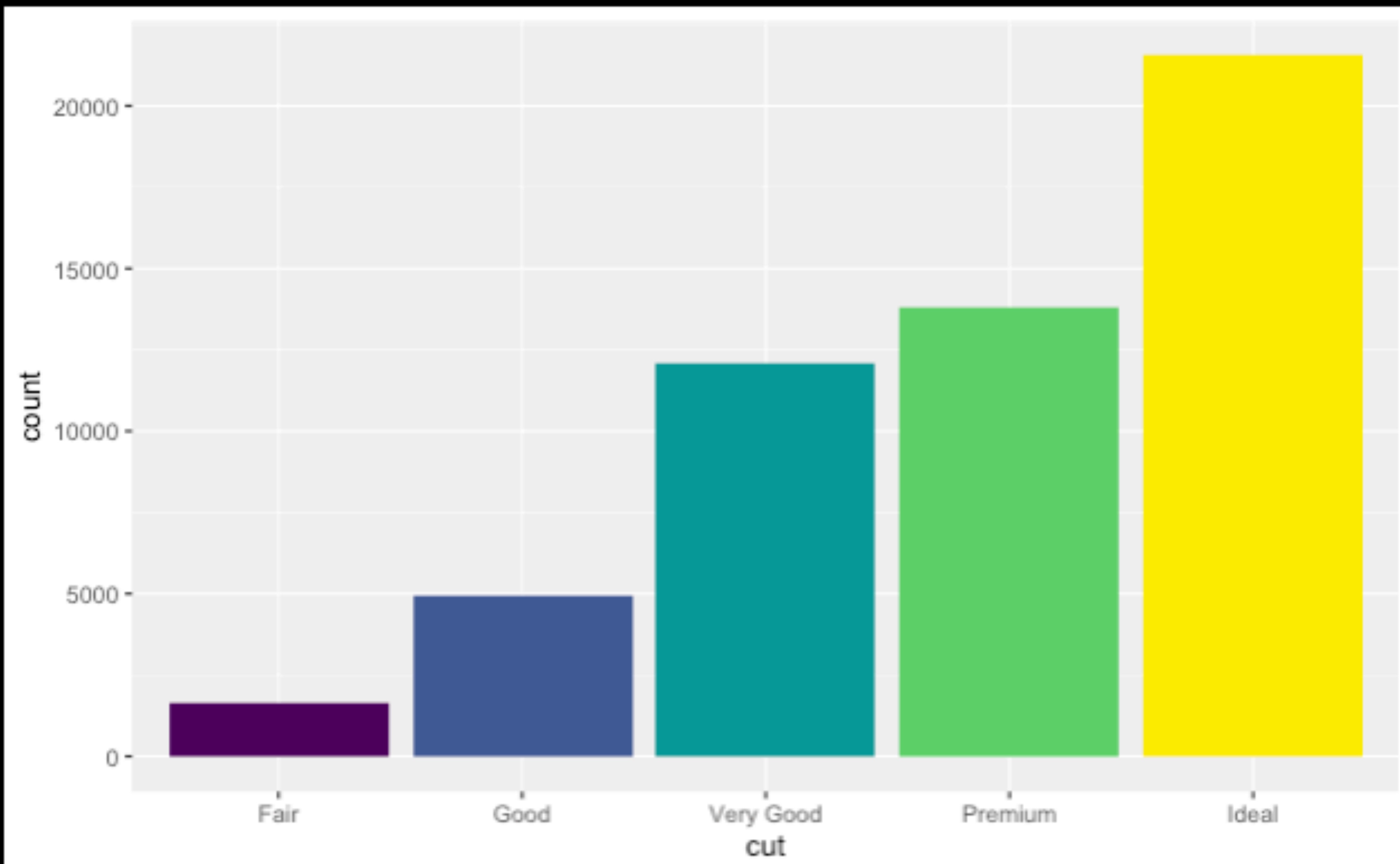


theme_void()
Pusty szablon,
widoczne są
tylko figury
geometryczne

```{r diamonds}

ggplot(data = diamonds) +
  geom_bar(
    mapping = aes(x = cut,
                  fill = cut),
    show.legend = FALSE)

```



1. Zacznijmy od zbioru danych **diamonds**

2. Za pomocą funkcji **stat_count** obliczmy, ile razy występuje każda wartość zmiennej **cut**

carat	cut	color	clarity	depth	table	price	x	y	z
0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31
0.29	Premium	I	VS2	62.4	58	334	4.20	4.23	2.63
0.31	Good	J	SI2	63.3	58	335	4.34	4.35	2.75
...	...	...	...	...	...	...	...	...	...

stat_count() →

cut	count	prop
Fair	1610	1
Good	4906	1
Very Good	12082	1
Premium	13791	1
Ideal	21551	1

3. Reprezentujemy każdą obserwację w wierszu

4. Mapujemy estetykę **`fill`** każdego wiersza na zmienną **`count`**

carat	cut	color	clarity	depth	table	price	x	y	z
0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31
0.29	Premium	I	VS2	62.4	58	334	4.20	4.23	2.63
0.31	Good	J	SI2	63.3	58	335	4.34	4.35	2.75
…	…	…	…	…	…	…	…	…	…

stat_count()

cut	count	prop
Fair	1610	1
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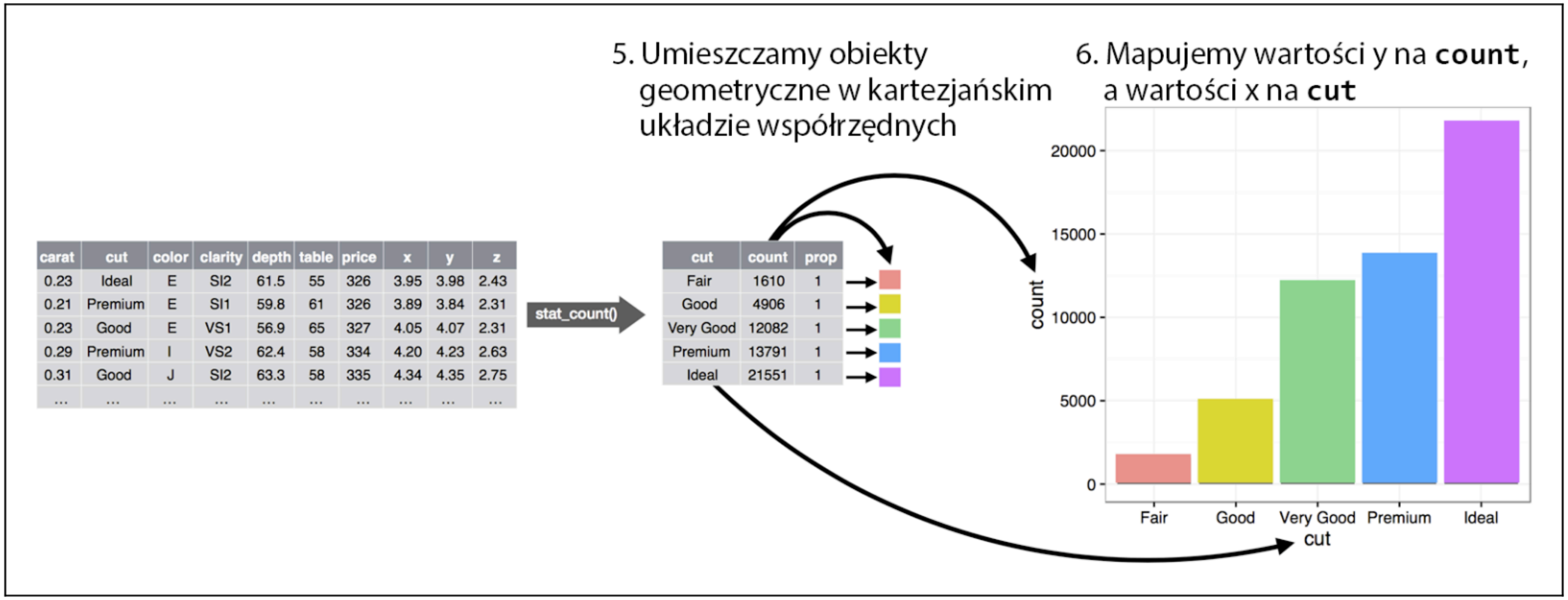
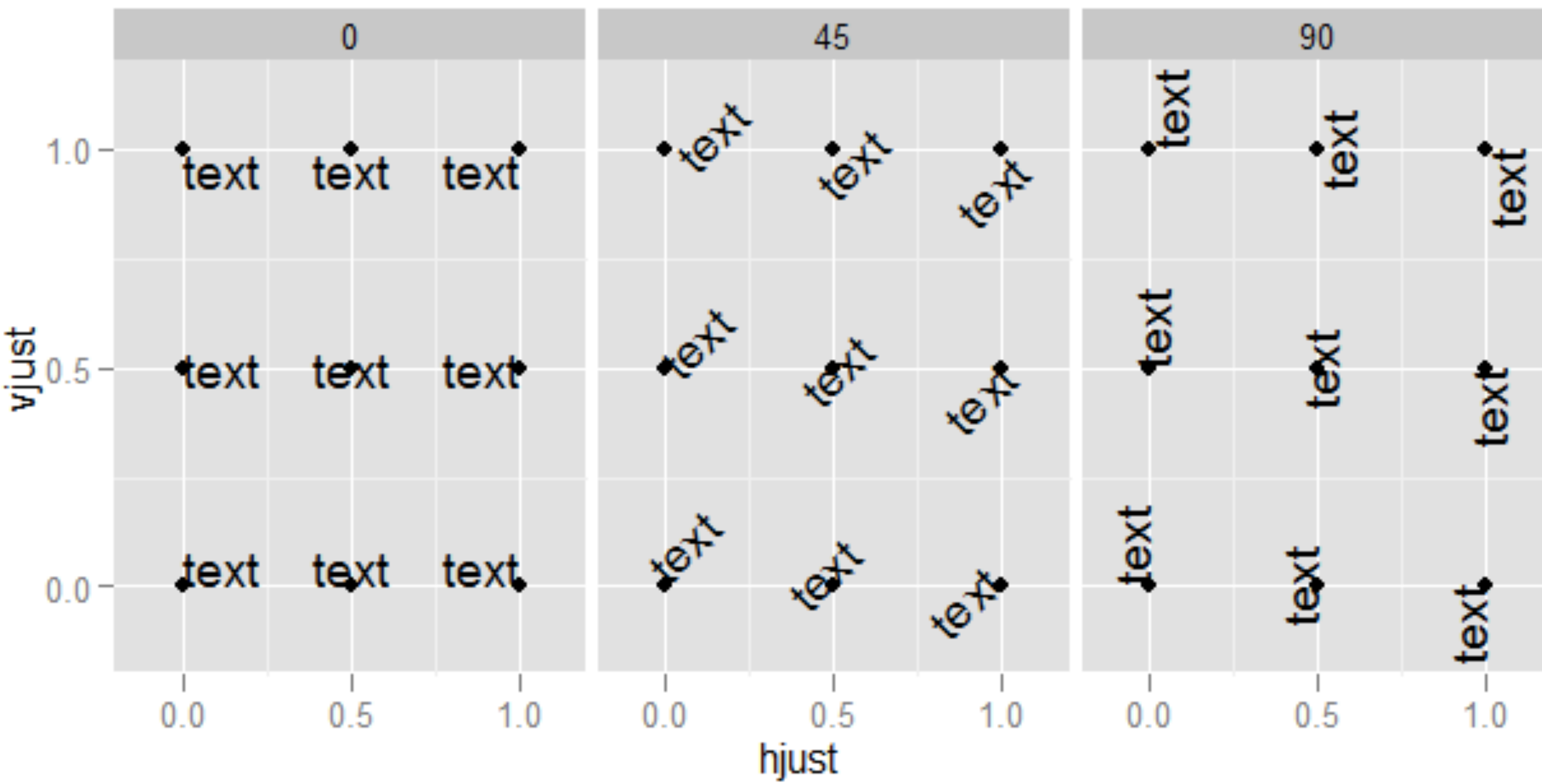
Premium	13791	1
Ideal	21551	1



5. Umieszczamy obiekty geometryczne w kartezjańskim układzie współrzędnych
6. Mapujemy wartości y na count, a wartości x na cut
carat cut color clarity depth table price x y z
0.23 Ideal E SI2 61.5 55 326 3.95 3.98 2.43
0.21 Premium E SI1 59.8 61 326 3.89 3.84 2.31
0.23 Good E VS1 56.9 65 327 4.05 4.07 2.31
0.29 Premium I VS2 62.4 58 334 4.20 4.23 2.63
0.31 Good J SI2 63.3 58 335 4.34 4.35 2.75
stat_count()
cut count prop
Fair 1610 1
Good 4906 1
Very Good 12082 1
Premium 13791 1
Ideal 21551 1
count
20000
15000
10000
5000
0
Fair
Good
Very Good
Premium
Ideal
cut

Operatory w R

Operator	Description
<	Less than
>	Greater than
<=	Less than or equal to
>=	Greater than or equal to
==	Equal to
!=	Not equal to
!x	Not x
x | y	x OR y
x & y	x AND y



0
45
90
text
vjust
hjust
1.0
0.5
0.0